

MCA with Specialisation in **Data Science**

Driving Tomorrow - The UPES Way

ADMISSIONS OPEN



ABOUT UPES ONLINE

UPES Online, the digital learning vertical of UPES, is aimed at empowering professionals looking to advance in their careers, students aspiring to upskill, and individuals striving to achieve personal goals. It offers a range of flexible, industry-relevant programs designed to equip learners with the skills needed to thrive in today's dynamic job market. UPES Online offers MBA, MCA, BBA, and BCA degrees, all enhanced with industry masterclasses and immersive experiences, maintaining the same level of academic rigour as traditional in-person programs. Additionally, the platform provides intensive 10-month Postgraduate Certification Programs (PGCP) in fields such as Oil and Gas, Renewable Energy, and Industrial Safety, with the integration of advanced technologies like AI, Data Science, and Cloud Computing into the curriculum.

UPES Online is an extension of UPES, a UGC-entitled, multidisciplinary, not-for-profit, research-driven university, committed to academic excellence, exemplary student outcomes, and social impact. As the 'University of Tomorrow', it pioneers transformative education, shaping learners into changemakers. With seven schools, including Computer Science, Engineering, Health Sciences and Technology, Design, Business, Law, Liberal Studies and Media, UPES provides industry-ready education to over 16,000+ students at its campuses in Dehradun.

As per the National Institutional Ranking Framework (NIRF) 2024, UPES has been ranked 46, with a rank of 42 in engineering and a rank of 41 in management. In addition to this, the university has been ranked the No.1 private university in academic reputation in India by the QS World University Rankings 2025 and is among the top 2% of universities in the world. UPES has also been accredited by NAAC with a grade 'A' and has received 5 stars on Employability (placements), by globally acclaimed QS Rating. The university has had 100% placements over the last five years. Forty-Six (46) faculty members of UPES are among the world's top 2% researchers as per Stanford University.

WHY UPES ONLINE?

**RANKED 46
IN INDIA**



**NO.1 PRIVATE UNIVERSITY
IN ACADEMIC REPUTATION**



**NAAC A
ACCREDITED**



**Legacy of
20+ Years**



**Runway Incubator
for Startup Support**



**Attractive
Scholarships**



**13 Research
Centers**



**Esteemed Faculty from
IITs, IISc and Ivy League**



**750+
Recruiters**



**Placement
Assistance**

ABOUT THE PROGRAM

The Master of Computer Applications (MCA) with a Specialization in Data Science is a postgraduate program that provides in-depth knowledge and advanced skills in computer science and data analytics, covering key areas such as software development, programming languages, algorithms, data structures, computer networks, databases, and software engineering. The program equips students to analyze complex data, design robust software systems, and manage applications across diverse domains like finance, healthcare, e-commerce, and education, preparing them to address the challenges of the digital age through technical expertise and data-driven decision-making.

KEY HIGHLIGHTS OF THE PROGRAM



Over 100 hours of self-paced tutorial videos featuring industry-oriented examples and case-study-based learning to enhance practical understanding.



Weekend live sessions conducted by seasoned academicians for in-depth learning.



Masterclasses and webinars by industry practitioners to bridge the gap between theory and practice.



A dedicated student support manager is assigned to guide and assist you throughout your academic journey.



Networking opportunities with peers to expand your professional connections and industry outreach.



Complimentary access to LinkedIn career essential programs to enhance your job readiness and career prospects.

ELIGIBILITY

- Passed any graduation degree (e.g.: B.E. / B.Tech./ B.Sc / B.Com. / B.A./ B. Voc./ BCA etc.,) preferably with Mathematics at 10+2 level or at Graduation level. Obtained at least 50% marks in the qualifying examination.
- For candidates with Non-Mathematics /IT/Computer Science background Candidates who have not studied Mathematics /IT/Computer Science in XII std. / Graduation, will be required to complete a Basic Mathematics/Basic Computer Information Systems course, respectively. Final admission will be subject to qualifying the aforesaid course within 6 weeks of provisional admission.
- Relaxation in Eligibility Criteria for Candidates with Work Experience
Candidates with minimum 2 years of work experience must have Bachelor's degree or equivalent in any discipline from a recognized university/institution with minimum 45 percent aggregate marks.



LEARNING OUTCOMES

- Demonstrate proficiency in programming languages like Python, Java, and C++ to design and implement efficient algorithms, data structures, and advanced applications that address real-world data science challenges.
- Utilise foundational and advanced machine learning models, deep learning algorithms, and statistical techniques to analyse, interpret, and derive insights from complex datasets.
- Employ cloud platforms and big data tools for processing, storing, and analysing large datasets, ensuring efficient and scalable data solutions in various domains.
- Create interactive and insightful visualisations using data visualisation tools and frameworks to effectively communicate data insights and facilitate informed decision-making.
- Showcase a strong understanding of ethical considerations, data privacy, and security in the context of data science, ensuring responsible use of data technologies to address industry-specific problems.

PROGRAM FEES

Indian Students

₹1,70,000

Duration

24 Months

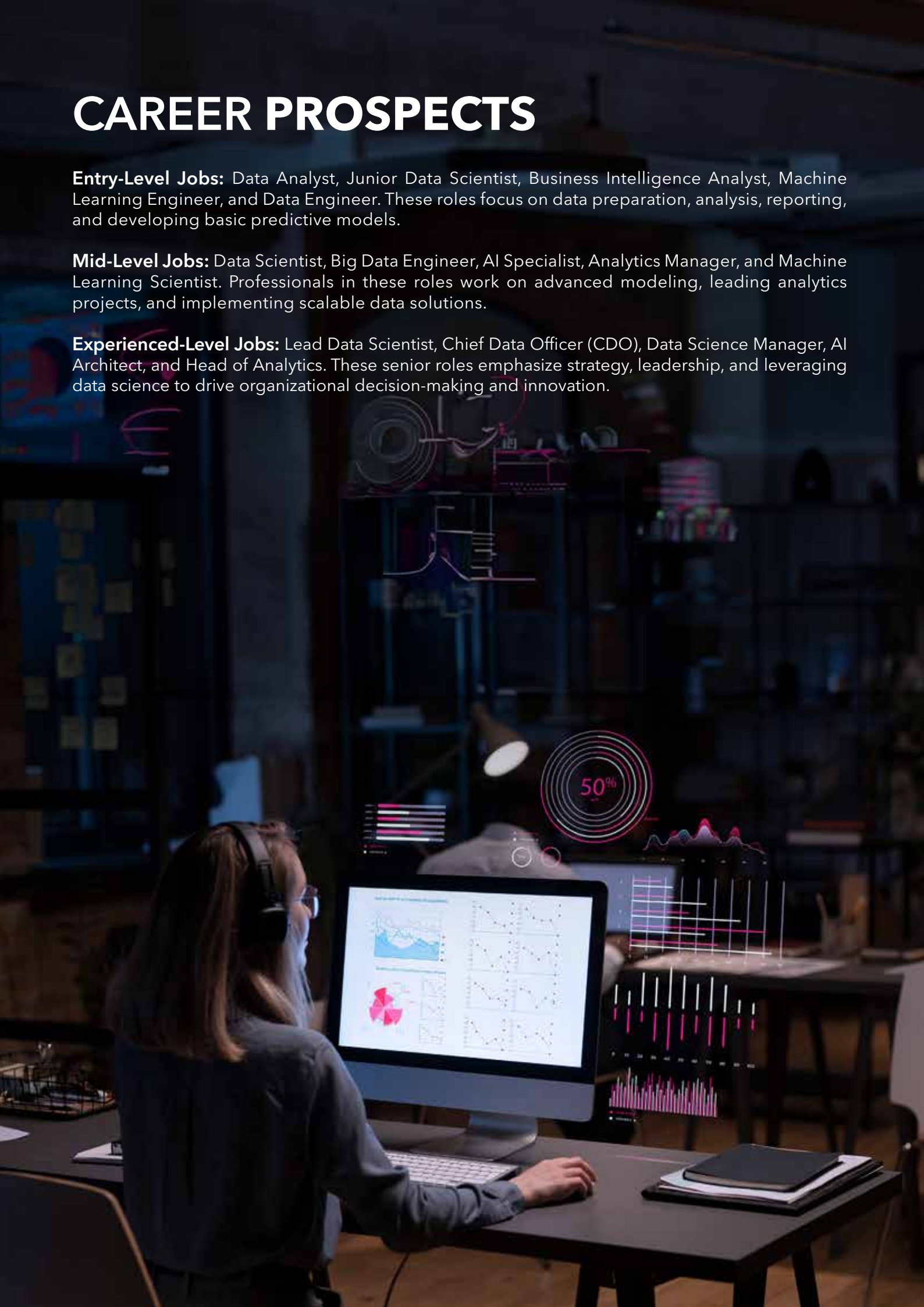
For FAQs - Visit upesonline.ac.in

CAREER PROSPECTS

Entry-Level Jobs: Data Analyst, Junior Data Scientist, Business Intelligence Analyst, Machine Learning Engineer, and Data Engineer. These roles focus on data preparation, analysis, reporting, and developing basic predictive models.

Mid-Level Jobs: Data Scientist, Big Data Engineer, AI Specialist, Analytics Manager, and Machine Learning Scientist. Professionals in these roles work on advanced modeling, leading analytics projects, and implementing scalable data solutions.

Experienced-Level Jobs: Lead Data Scientist, Chief Data Officer (CDO), Data Science Manager, AI Architect, and Head of Analytics. These senior roles emphasize strategy, leadership, and leveraging data science to drive organizational decision-making and innovation.



CURRICULUM

Semester 1

- Advance Python Programming
- Data Base Management Systems
- Discrete Mathematics
- Object Oriented Programming using C++
- Advance Data Structures
- Fundamentals of Data Science

Semester 3

- Object-Oriented Analysis and Design Using UML
- Computer Networks
- Advanced Java Programming
- Software Engineering and Project Management
- AR/VR Development
- Machine Learning and Deep Learning

Semester 2

- Web Technologies
- Operating Systems
- Design and Analysis of Algorithm
- Java Programming
- Data Visualization and Interpretation
- Computer Graphics

Semester 4

- Project Dissertation
- Android Application Development
- Cloud Computing for Data Science
- Use of cloud platforms for data processing analysis and storage
- Generative Artificial Intelligence



CAREER SERVICES

Our comprehensive career services guarantee your career preparedness and employability for the professional world.

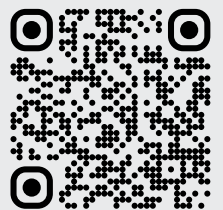
- Career Orientation
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RECRUITERS



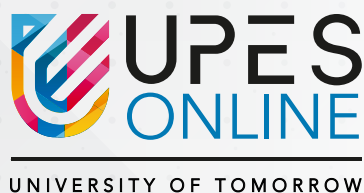


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MCA with Specialisation in

Cyber Security and Forensics

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ABOUT THE PROGRAM

The global cybersecurity market is projected to grow at a CAGR of 13.4%, reaching \$376.3 billion by 2028, driven by the increasing need to protect data and digital infrastructure from cyber threats. As industries like finance, healthcare, and IT face rising risks of cyberattacks, the demand for cybersecurity experts continues to soar, making it one of the most sought-after and high-paying career paths.

UPES Online's 2-year MCA in Cyber Security and Forensics equips students with cutting-edge skills in ethical hacking, digital forensics, malware analysis, and network security. The program focuses on emerging areas like blockchain security, cloud security, and penetration testing, ensuring learners are ready to address the challenges of modern cyber threats. With a curriculum blending technical expertise and real-world problem-solving, graduates are well-prepared to secure critical digital assets across industries.

KEY HIGHLIGHTS OF THE PROGRAM



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Candidates with minimum 2 years of work experience must have Bachelor's degree or equivalent in any discipline from a recognized university/institution with minimum 45 percent aggregate marks.



LEARNING OUTCOMES

- Utilise advanced programming languages like Python, Java, and C++ alongside algorithm design and data structures to create robust and efficient solutions addressing real-world cybersecurity challenges.
- Develop secure architectures using cryptography, network security protocols, and advanced technologies such as ethical hacking and penetration testing to protect systems against cyber threats and vulnerabilities.
- Apply digital forensic methods to investigate cyber incidents, recover digital evidence, and provide actionable insights for threat mitigation while adhering to legal and regulatory standards.
- Leverage tools such as AR/VR, Android application development, and web technologies to enhance cybersecurity frameworks, ensuring secure and seamless integration with modern IT infrastructures.
- Exhibit a deep understanding of ethical principles and professional practices in cybersecurity and forensics, ensuring responsible use of technology to safeguard privacy, data, and digital assets.

PROGRAM FEES

Indian Students

₹1,70,000

Duration

24 Months

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CAREER PROSPECTS

Entry-Level Cyber Security and Forensics:

At the entry level, critical success factors focus on basic operational efficiency, customer satisfaction, quality control, timely delivery, and cost management. These ensure smooth day-to-day operations and a reliable product or service.

Mid-Level Cyber Security and Forensics:

At the mid-level, the focus shifts to growth and competitiveness through innovation, talent management, brand recognition, customer retention, and market expansion. These factors help scale the business and improve its market position.

Experienced-Level Cyber Security and Forensics:

At the experienced level, critical success factors revolve around strategic leadership, building partnerships, data-driven decision-making, strong organizational culture, and leveraging technology for a competitive edge. These drive long-term success and market leadership.



CURRICULUM

Semester 1

- Advance Python Programming
- Data Base Management Systems
- Discrete Mathematics
- Object Oriented Programming using C++
- Advance Data Structures
- Information Technology and Cyber Security

Semester 3

- Object-Oriented Analysis and Design Using UML
- Computer Networks
- Advanced Java Programming
- Software Engineering and Project Management
- AR/VR Development
- Digital Forensics- I

Semester 2

- Web Technologies
- Operating Systems
- Design and Analysis of Algorithm
- Java Programming
- Cryptography & Network Security
- Computer Graphics

Semester 4

- Project Dissertation
- Android Application Development
- Digital Forensics- II,
- Ethical Hacking & Penetration Testing



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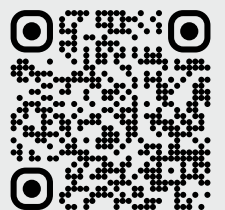
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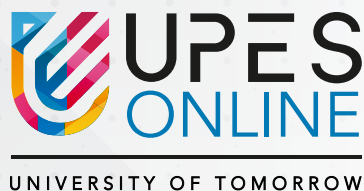


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ABOUT THE PROGRAM

The global artificial intelligence market is expected to grow at a staggering CAGR of 37.3%, reaching \$1.81 trillion by 2030, fueled by its transformative impact on industries like healthcare, automotive, retail, and finance. With AI-driven innovations reshaping businesses and processes, the demand for professionals skilled in AI and machine learning is at an all-time high, offering some of the most rewarding career opportunities.

UPES Online's 2-year MCA in Artificial Intelligence and Machine Learning prepares students with expertise in deep learning, natural language processing (NLP), computer vision, and AI model deployment. The program emphasizes practical skills in developing AI solutions, leveraging neural networks, and implementing scalable machine learning algorithms. With a curriculum designed to meet the needs of AI-driven industries, graduates are empowered to excel in roles like AI Engineer, Data Scientist, and Machine Learning Specialist, contributing to the future of intelligent systems.



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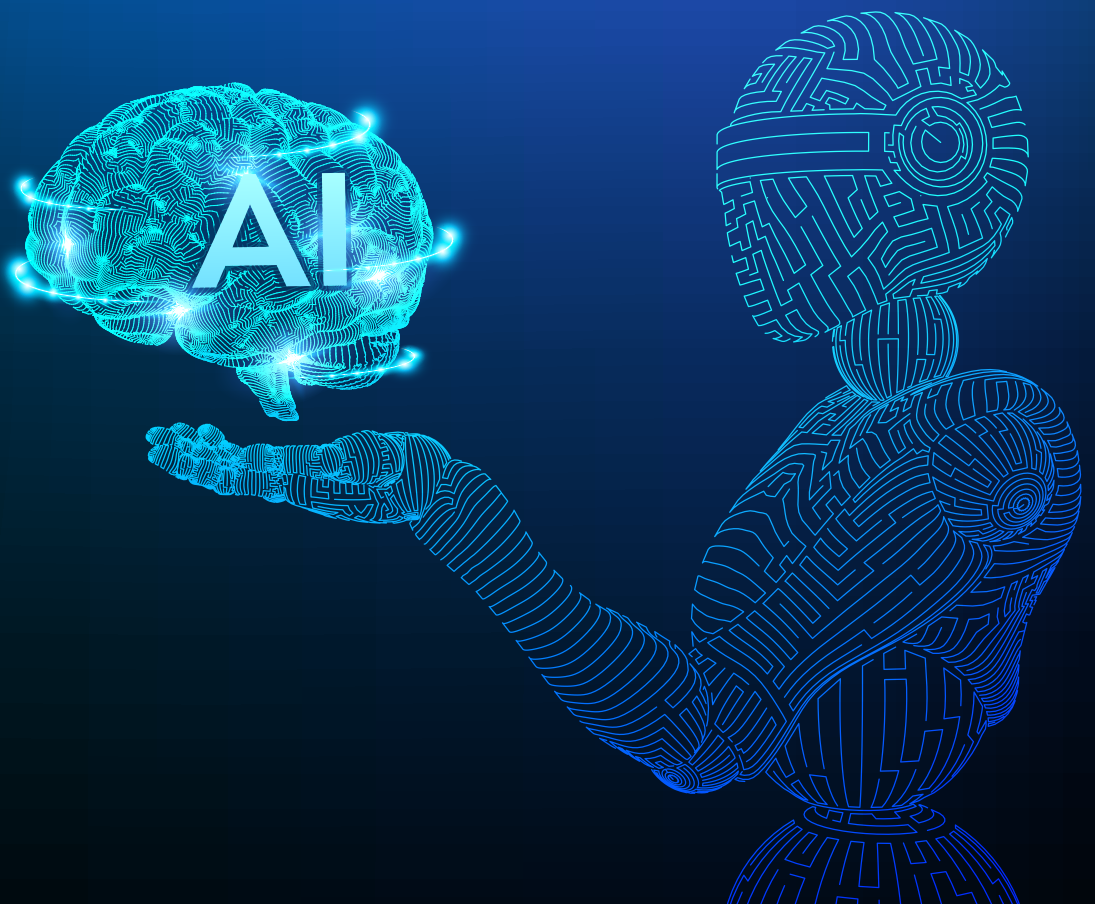
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LEARNING OUTCOMES

- Develop and implement efficient algorithms using advanced programming languages like Python, Java, and C++ while leveraging data structures to address complex computational challenges.
- Analyse and apply artificial intelligence and machine learning concepts, including neural networks, deep learning, and natural language processing, to solve domain-specific problems in diverse industries.
- Utilise tools and frameworks such as web technologies, advanced Java programming, and object-orientated design to create scalable and secure web and mobile applications.
- Employ mathematical and statistical foundations, such as discrete mathematics, data analysis, and machine learning techniques, to design and optimise AI/ML models with high accuracy.
- Integrate emerging technologies like AR/VR, pattern recognition, and Android development into intelligent systems while adhering to ethical guidelines and ensuring the privacy, security, and fairness of AI solutions.

PROGRAM FEES

Indian Students

₹1,70,000

Duration

24 Months

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CAREER PROSPECTS

Entry-Level Artificial Intelligence and Machine Learning Roles:

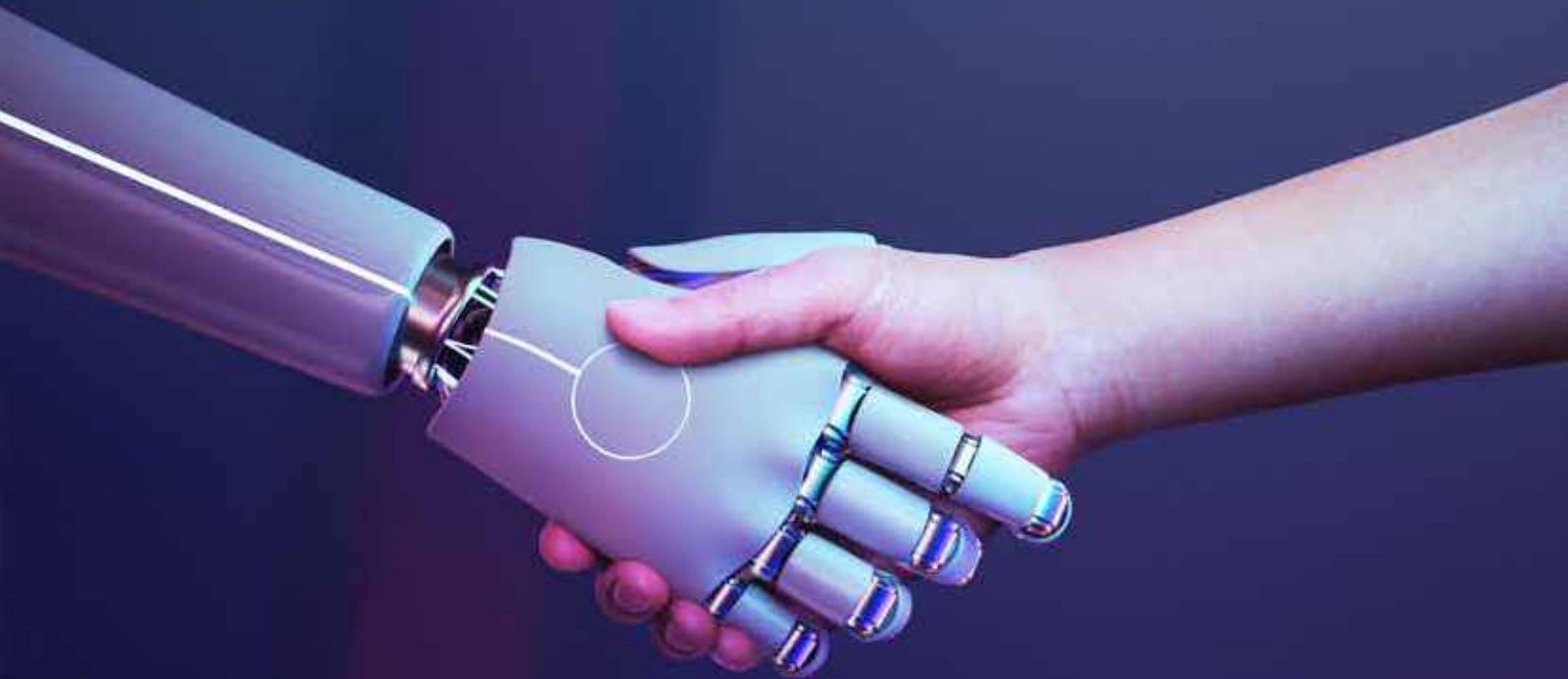
At the entry level, AI/ML professionals focus on foundational tasks like data preprocessing, building simple models, and analyzing data. They work with basic machine learning algorithms and are responsible for cleaning data and extracting insights. Roles at this stage include Junior Data Scientist, Machine Learning Engineer, and Data Analyst.

Mid-Level Artificial Intelligence and Machine Learning Roles:

Mid-level AI/ML professionals handle more complex tasks such as developing scalable models, implementing advanced algorithms, and deploying models into production. They lead smaller projects, collaborate with cross-functional teams, and optimize models for real-world applications. Roles include Data Scientist, AI Engineer, and Machine Learning Specialist.

Experienced-Level Artificial Intelligence and Machine Learning Roles:

Experienced AI/ML professionals focus on leadership and innovation, overseeing complex AI systems and driving strategic AI initiatives across the organization. They design AI systems, lead teams, and ensure machine learning solutions meet business objectives. Roles at this stage include AI Architect, Lead Data Scientist, and Chief AI Officer.



CURRICULUM

Semester 1

- Advance Python Programming
- Data Base Management Systems
- Discrete Mathematics
- Object Oriented Programming using C++
- Advance Data Structures
- AI for everyone

Semester 3

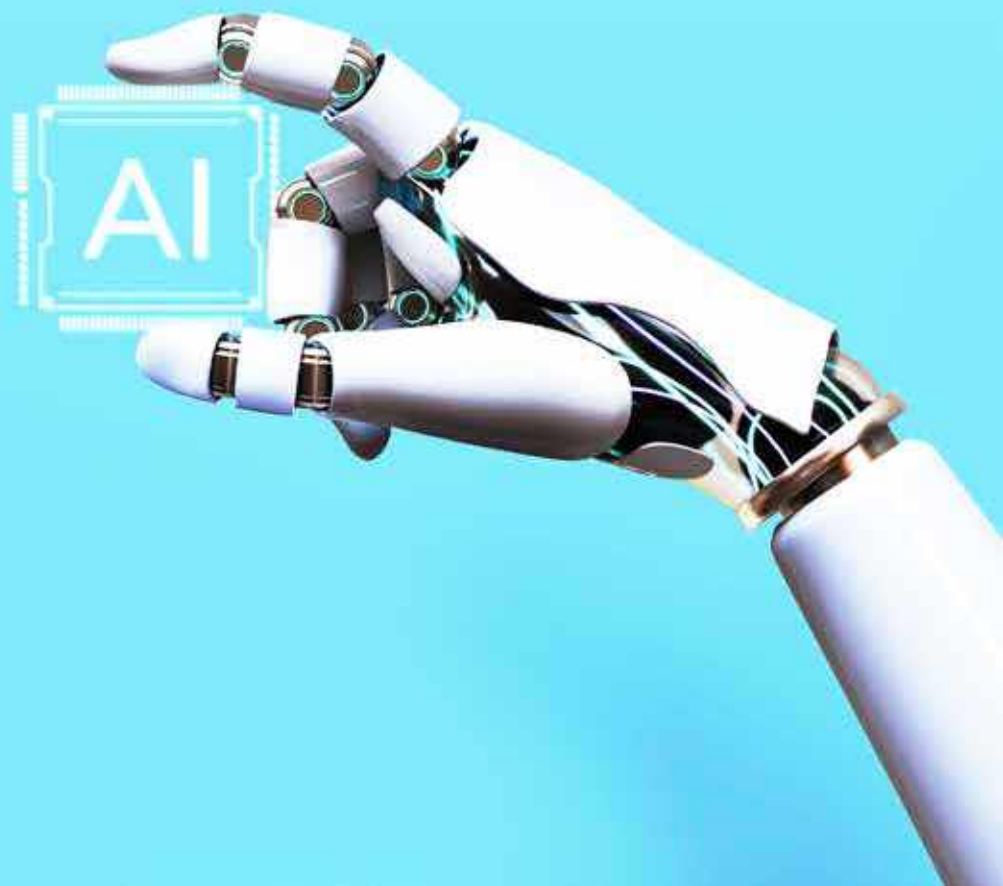
- Object-Oriented Analysis and Design Using UML
- Computer Networks
- Advanced Java Programming
- Software Engineering and Project Management
- AR/VR Development
- Deep Learning

Semester 2

- Web Technologies
- Operating Systems
- Design and Analysis of Algorithm
- Java Programming
- Applied Machine Learning
- Computer Graphics

Semester 4

- Project Dissertation
- Android Application Development
- Pattern and Visual Recognition
- Computational Linguistics and Natural Language Processing



CAREER SERVICES

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- Career Orientation
- LinkedIn/Resume building guidance
- Mentorship by industry experts
- Interview preparation

RECRUITERS

amazon

Coca-Cola

mahindra

KPMG

TATA
TATA POWER

LARSEN & TOUBRO

Adecco

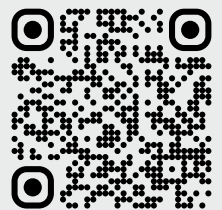
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